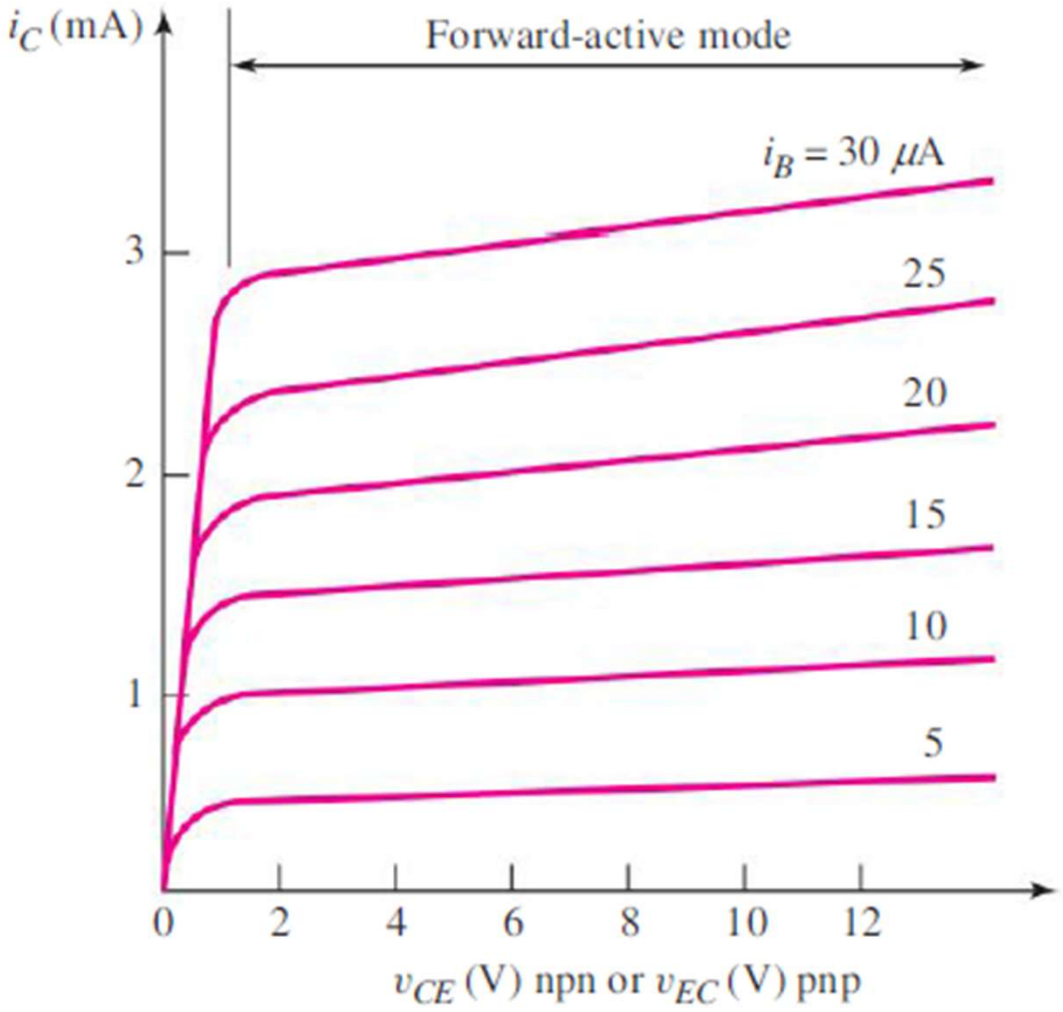
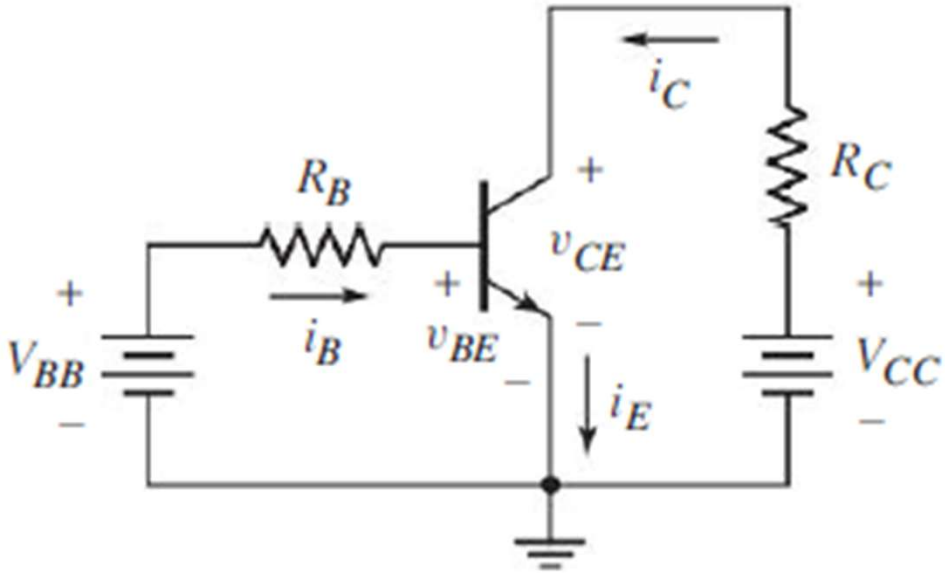
Electronic Circuits and Applications

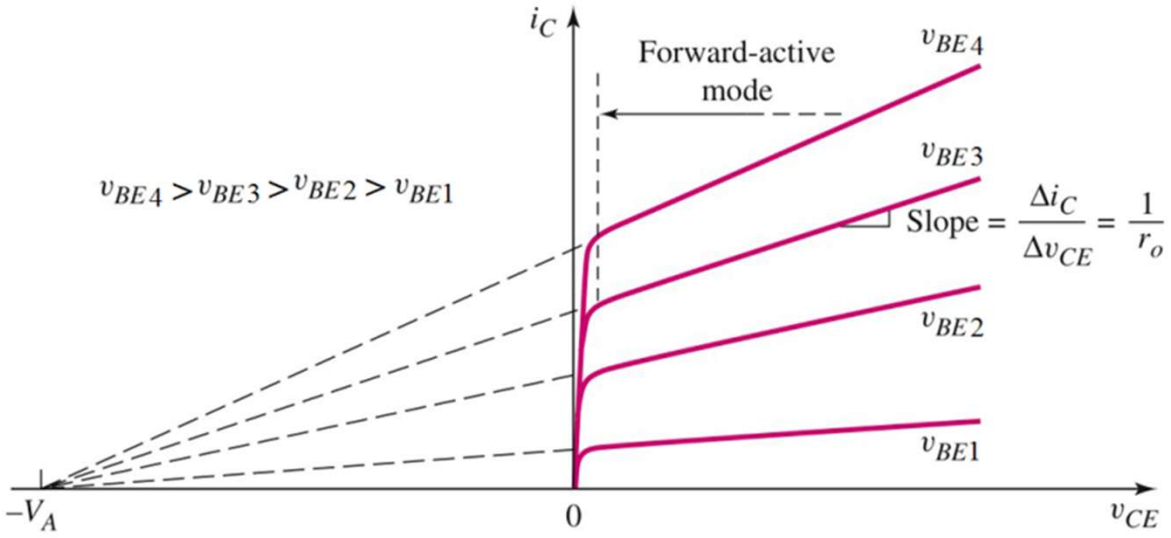
Characteristics of CE BJT





V_{BB} source forward biases the B–E junction and controls the base current i_B. The C–E voltage can be varied by changing V_{CC}.

Early Effect



$$r_0 \cong \frac{V_A}{I_C}$$

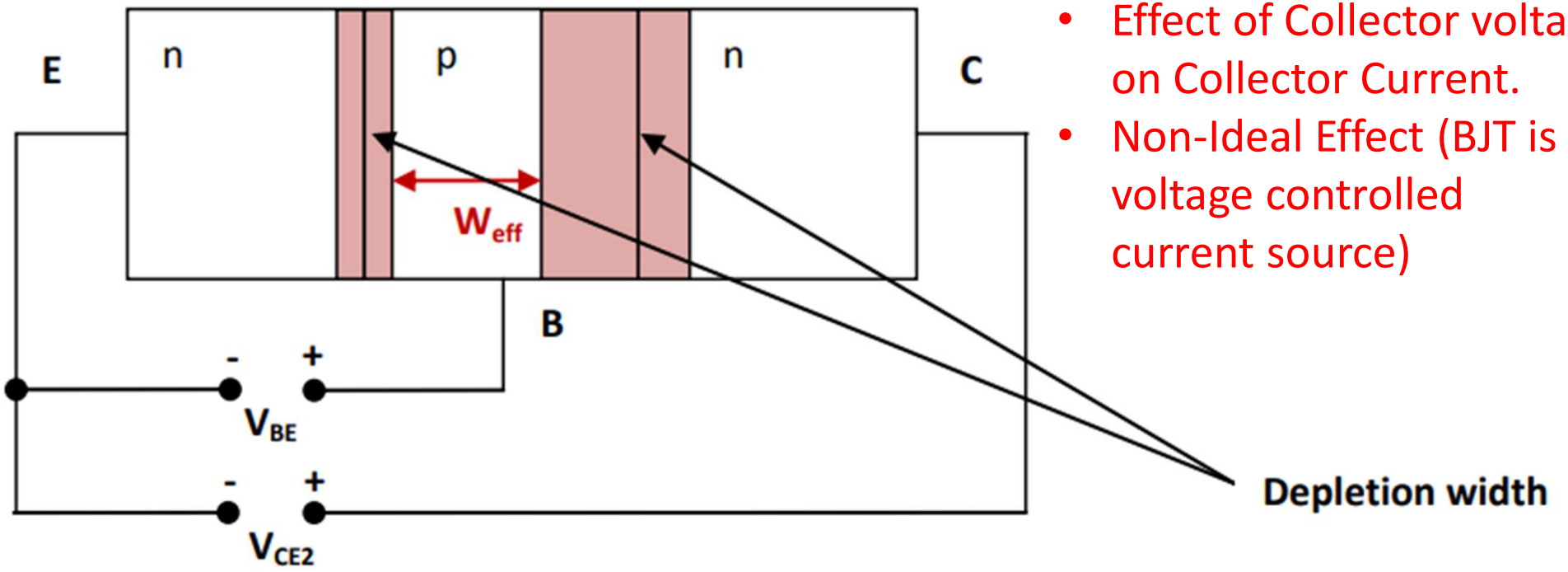
NOTE: V_{EC} for PNP

I-V characteristic of common-emitter BJT circuit, showing Early voltage and the finite output resistance, of the transistor

The early effect is the variation in the width of the base in a BJT due to a variation in the applied base-to-collector voltage.

For example a greater reverse bias across the collector- base junction increases the collector-base depletion width.

Early Effect



- Effect of Collector voltage on Collector Current.
- Non-Ideal Effect (BJT is voltage controlled current source)

Early voltage is the voltage which when externally applied to BJT as Vce will make Base width extremely thin (zero for practical purposes). So any change in input voltage V_{BE} will fail to change the output current Ic.

Base-width modulation (Early effect)

For constant V_{BE} , V_{CE} ↑ ---- V_{CB} ↑

Neutral base width W ↓

Minority carrier concentration gradient ↑

------- IC ↑

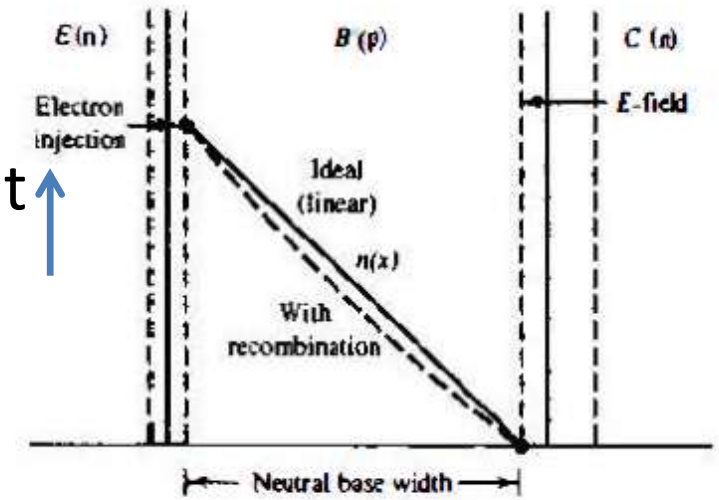


Figure 3.4 Minority carrier electron concentration across the base region of an npn bipolar transistor biased in the forward-active mode

For a given value of v_{BE} in an npn transistor, if v_{CE} increases, the reverse-bias voltage on the collector– base junction increases, which means that the width of the B–C space-charge region also increases. This in turn reduces the neutral base width W. A decrease in the base width causes the gradient in the minority carrier concentration to increase, which increases the diffusion current through the base. The collector current then increases as the C–E voltage increases.

Early Effect

Consequences of Reduced Base Width:

Increased Collector Current (I_C): A narrower base means that minority carriers (electrons in a PNP, holes in an NPN) injected from the emitter into the base have a shorter distance to travel to reach the collector. This reduces the chances of recombination within the base, leading to a higher percentage of carriers reaching the collector, and thus an increase in collector current (I_C).

Reduced Base Current (I_B): Since fewer carriers recombine in the base, the base current (which primarily supplies these recombining carriers) decreases.

Output Resistance: The change in collector current with collector-emitter voltage (V_{CE}) due to the Early effect means that the BJT has a finite output resistance in the active region, rather than being an ideal current source.

Early Voltage (V_A): If you extrapolate the linear part of the I_C vs. V_{CE} curves back to the voltage axis, they all intersect at a common point called the Early voltage (V_A). The Early voltage is a measure of how strongly the collector current is affected by the Early effect. A higher Early voltage indicates a less pronounced Early effect and a more ideal current source behavior.

Early Effect

- Effect of Collector voltage on Collector Current.
- Non-Ideal Effect (BJT is voltage controlled current source)

The Early effect describes how the collector current of a BJT is influenced by the collector-emitter voltage due to the modulation of the effective base width. This effect is important for understanding the AC characteristics and output impedance of BJTs, particularly in amplifier design.

The linear dependence of i_C versus v_{CE} in the forward-active mode can be described by

$$i_C = I_S(e^{v_{BE}/V_T}) \cdot \left(1 + \frac{v_{CE}}{V_A}\right)$$

$$\frac{1}{r_o} = \left.\frac{\partial i_C}{\partial v_{CE}}\right|_{v_{BE}=\text{const.}}$$

$$r_o \cong \frac{V_A}{I_C}$$

Where I_C is the quiescent collector current when v_{BE} is a constant and v_{CE} is small compared to V_A.

STABILITY FACTORS

❖ **Stability Factor *S*:-** The stability factor *S, as the change of collector current with respect to the reverse saturation* current, keeping β *and* V_{BE} *constant. This can be written as:*

$$S \equiv \frac{\partial I_C}{\partial I_{CO}} \qquad \text{Or,} \qquad S = (1+\beta)\frac{1 + R_b/R_e}{1 + \beta + R_b/R_e}$$

❖ **Stability Factor *S'*:-** The variation of I_C *with* V_{BE} *is given by the stability factor S defined by the partial derivative:*

$$S' \equiv \frac{\partial I_C}{\partial V_{BE}} \approx \frac{\Delta I_C}{\Delta V_{BE}}$$

❖ **Stability Factor *S"*:-** The variation of I_C *with respect to* β *is represented by the stability factor, S'', given as:*

$$S'' \equiv \frac{\partial I_C}{\partial \beta} \approx \frac{\Delta I_C}{\Delta \beta}$$

Bias stabilization.

- A transistor can work as amplifier, only if the dc/ac voltages and currents in the circuit are suitably fixed. The operating point or **bias** point or quiescent point(or simply Q-point) is the voltage or current which, when applied to a device, causes it to operate in a certain desired fashion. Need for **bias stabilization**.

Need for BIAS STABILIZATION

- After fixing the operating point suitably, it should remains there only. But there are two reasons for the operating point to shift.
- The transistor parameters such as VBE & β changes from device to device
- Transistor parameters are also temperature dependent. Since the collector current is $I_C = \beta I_B + (1+\beta) I_{CO}$ (β, I_B and I_{CO} are temperature dependent)
- β of a transistor is strongly dependent on temperature. As temp. increases, β increases

- The general expression of stability factor for a CE configuration can be obtained as under.

$$I_c = \beta I_B + (1 + \beta) I_{c0}$$

- Differentiating above expression with respect to I_C, we get

$$S = \frac{1 + \beta}{1 - \beta \dfrac{dIB}{dIc}}$$

- Hence the stability factor S depends on β, I_B and I_C.

Stability factor- Voltage divider bias

To determine stability factor, S Applying Thevenin's Theorem to the circuit

$$V_T = \frac{R_2 V_{CC}}{R_1 + R_2} \; and R_B = \frac{R_1 R_2}{R_1 + R_2}$$

The loop equation around the base circuit can be written as

$$V_T = I_B R_B + V_{BE} + (I_B + I_C) R_E$$

Differentiating this equation wrt Ic, we get

$$\frac{dI_B}{dI_C} = -\frac{R_E}{R_E + R_B}$$

$$S = \frac{1 + \beta}{1 + \beta \dfrac{R_E}{R_E + R_B}}$$

- Example: In a CE ge transistor amplifier circuit, the bias is provided by bias, i.e. emitter resistor and potential divider arrangement. The various parameters are Vcc=16V, Rc=3K,Re=2K,R1=56K,R2=20K and α=0.985. Determine (a) the coordinates of the operating point and (b) the stability factor S.

Solution: For a ge transistor Vbe=0.3v, as α=0.985

(a) To find coordinates of the operating point

$$\beta = \frac{\alpha}{1-\alpha} = \frac{0.985}{1-0.985} = 66$$

$$R_B = \frac{R_2}{R_1 + R_2} . V_{cc} = \frac{20X10^3}{76X10^3} .16 = 4.21V$$

$$R_B = \frac{R_2}{R_1 + R_2} = \frac{20X10^3}{76X10^3} = 14.737K\Omega$$

- The loop equation around the base circuit is

$$V_T = I_B R_B + V_{BE} + (I_B + I_C) R_E$$

$$3.91 = I_C [0.223 + 2.03].10^3$$

$$I_c = 1.73 mA$$

$$V_{CE} = V_{CC} - I_c R_c - I_E R_E = 7.35V$$

- (b) To find the stability factor S,

$$S = (1+\beta).\frac{1+\frac{R_B}{R_E}}{1+\beta+\frac{R_B}{R_E}}$$

$$S = (1+66).\frac{1+\frac{14.737}{2}}{1+66+\frac{14.737}{2}} = 7.536$$

What is Thermal Runaway?

A self-accelerating, destructive process in BJTs caused by a positive feedback loop between temperature and current.

The Vicious Cycle

Temperature Increase (↑T): Initial heat from operation or ambient conditions.

Parameter Changes:

↑β **(Current Gain):** Due to increased minority carrier lifetime.

↑I_{CO} **(Leakage Current):** Doubles approximately every 10∘C.

↓V_{BE} **(Base-Emitter Voltage):** Reduces for a given current, potentially increasing I_B.

- These changes cause a significant **increase in Collector Current (↑I_C)**, leading to higher **Power Dissipation (↑P_D=V_{CE}×I_C)**.
- Increased P_D generates more heat, further escalating the temperature, completing the positive feedback loop.

Consequences:

Junction temperature exceeds maximum ratings (e.g., 150−200∘C).

Irreversible material degradation (melting, doping changes).

Device failure.

Key Prevention Strategies:

Proper Biasing: e.g., voltage divider bias to stabilize IC.

Heat Sinks: Efficiently dissipate heat to the environment.

Derating: Operating below maximum power specifications.

Thermal Compensation/Protection: Using temperature-sensitive components or shutdown circuits.

Thermal runaway

- The collector current for the CE circuit is given by

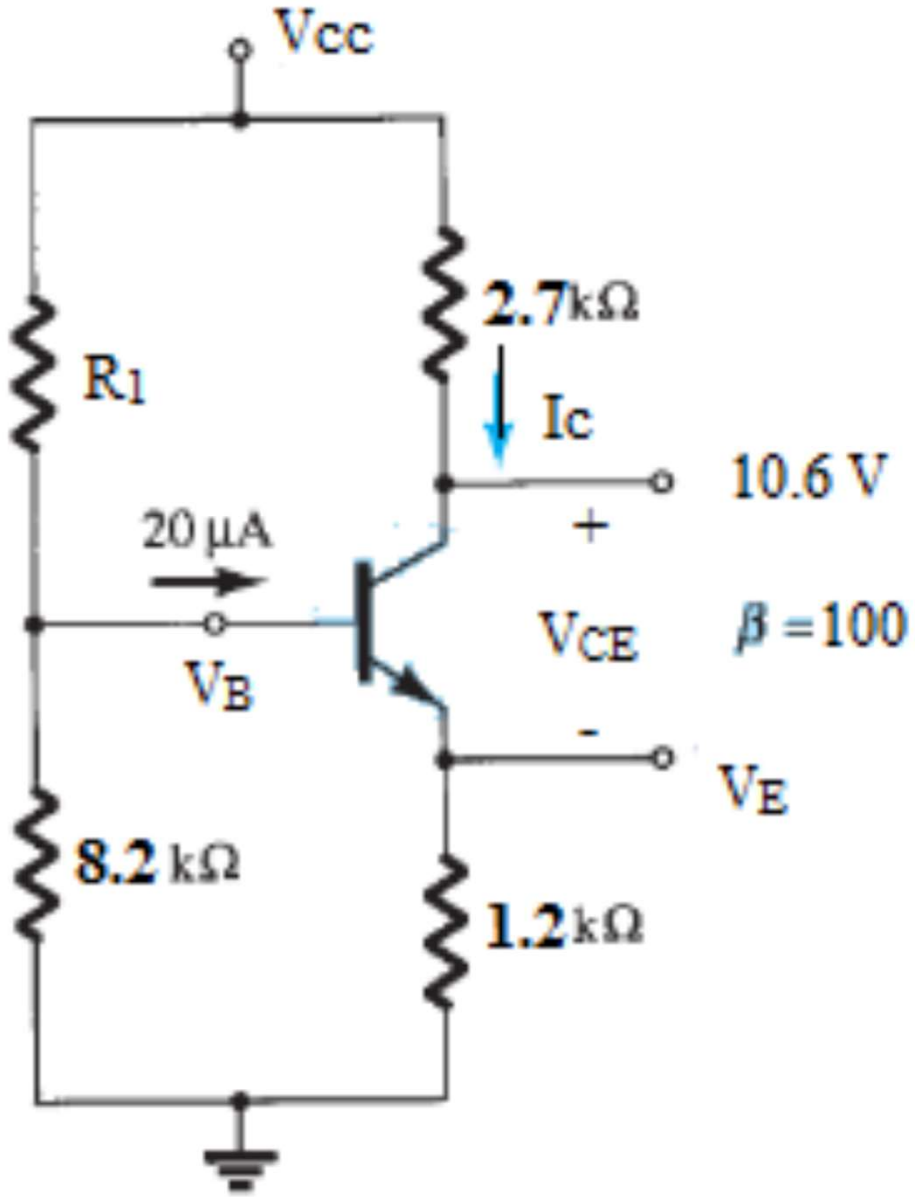
$$I_c = \beta I_B + (1 + \beta) I_{c0}$$

- The three variables in the equation, β ,IB and IC0 increase with rise in temperature.
- In particular, the reverse saturation current changes greatly with temperature.

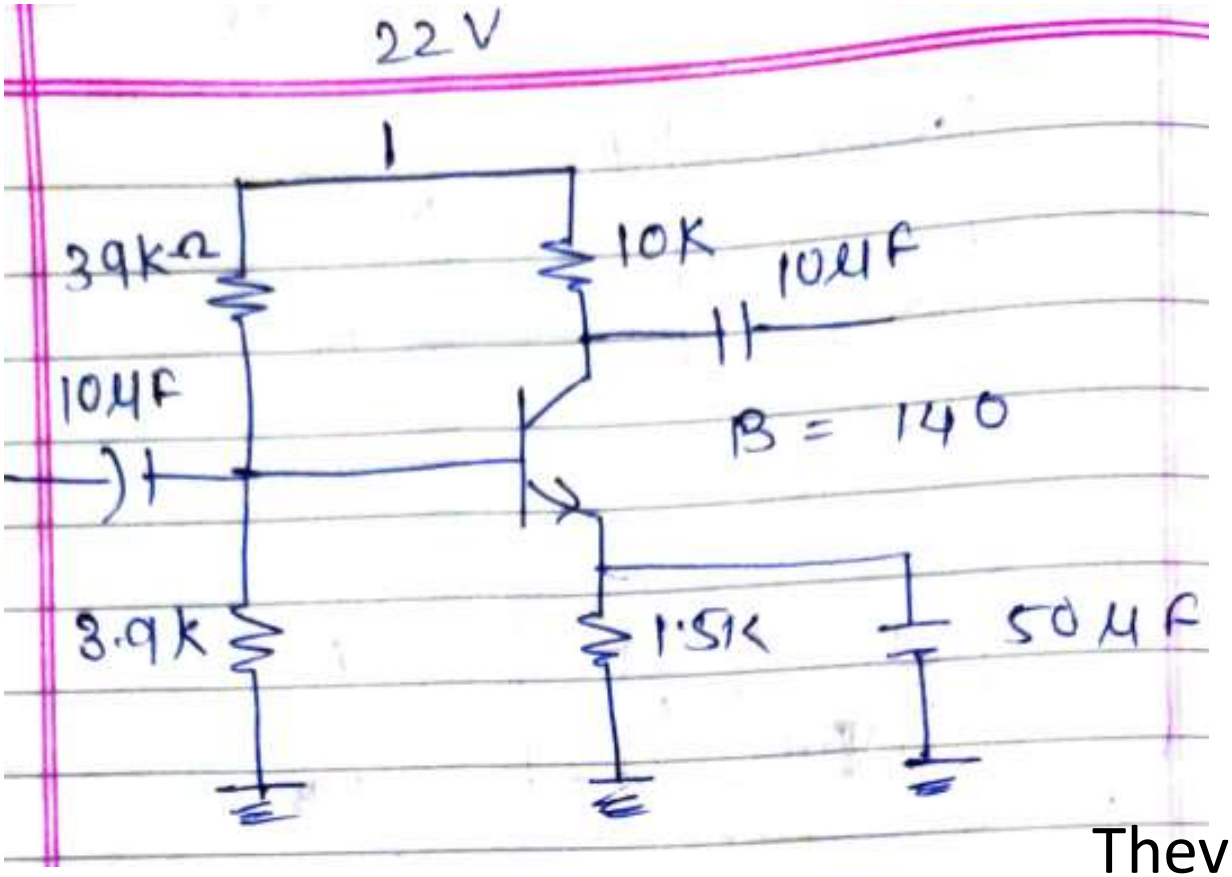
- The collector current causes the collector base junction temperature to rise which, in turn, increase leakage current, as result Ic will increase still further, which will further rise the temperature at the collector base junction.

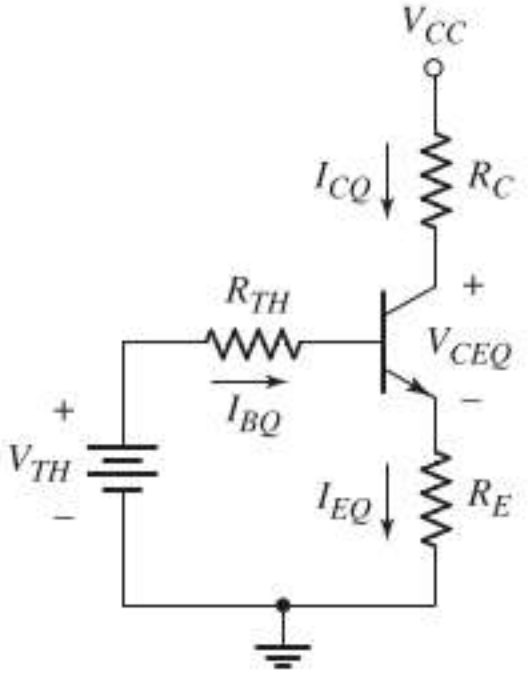
- This process will become cumulative leading to "**thermal runaway**".

Determine the following Parameters for the given circuits



A. I_C
B. V_{CE}
C. V_{CC}
D. V_E
E. V_B
F. R_1





Calculate operating point.

Thevenise the circuit

Vth= 1.999V Rth = 3.55K ohm

Apply KVL around BE loop

Calculate **I_B = 6.05 Micro A**

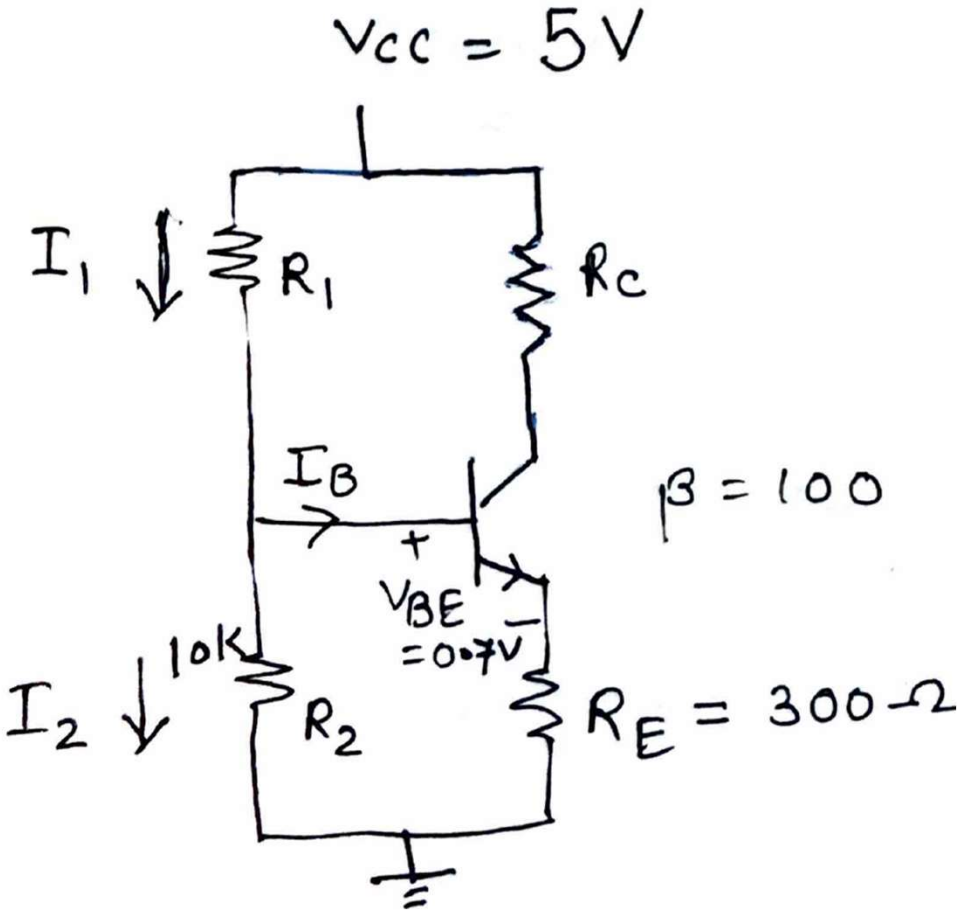
Ic = β I_B =

Apply KVL around CE loop

Calculate V_{CE}

V_{CE} =

Calculate R1, Rc, Ic = 1 mA, V_{CE}= 2.5 V



$R_1 = V_{R1} / I_1$

$= (5 - V_B) / (I_2 + I_B)$

I_B = 10 micro A

$I_E = 101\ I_B$

$= 1.01$ mA

$V_{RE} = I_E R_E$

$= 0.303V$

$V_B = V_{BE} + V_E$

$= 1.003V$

$= I_2 R_2$ Calculate I_2

$I_2 = 0.1003$ mA

$I_1 = I_B + I_2$

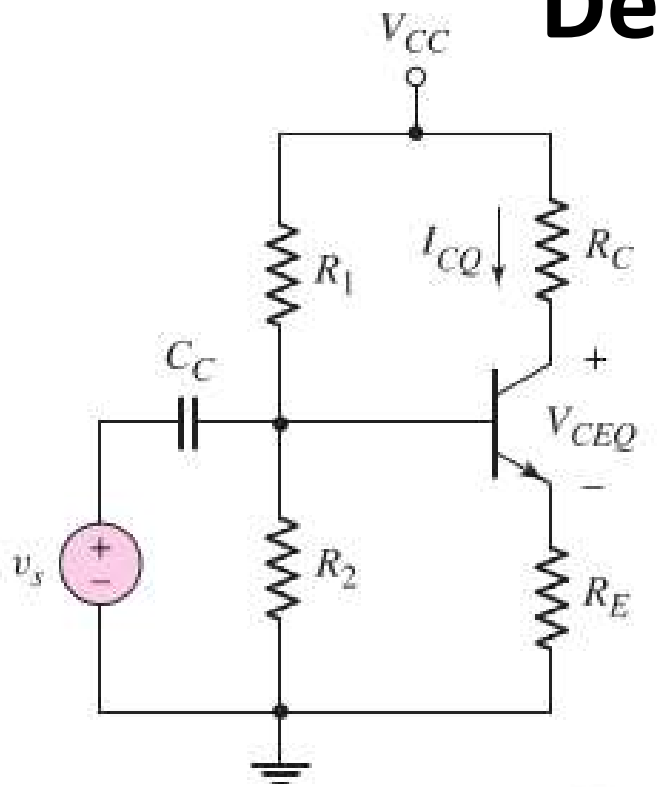
$= 0.1103$ mA

Calculate R_1

$R_1 = (5 - 1.003) / 0.1103$

$= 36.24$ K

Design bias stable circuit



V_{CC} = 5 V; $R_C = 1K$

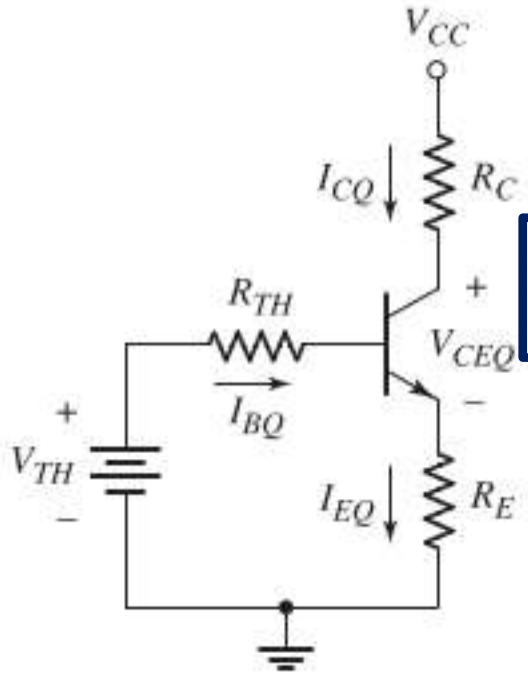
Choose R_E and determine R_1, R_2 such that circuit is considered bias stable.

V_{CE} = 3 V; $\beta = 120$; V_{BE} = 0.7 V

Assume $I_{CQ} = I_{EQ}$

Calculate I_{CQ}

Calculate I_{BQ}



For a bias-stable circuit, $R_{TH} = 0.1(1+\beta)R_E$

Calculate R_{TH}

Calculate R1 & R2

Given Data

- $V_{CC} = 5\ \text{V}$
- $R_C = 1\ \text{k}\Omega = 1000\ \Omega$
- $V_{CEQ} = 3\ \text{V}$
- $\beta = 120$
- $V_{BE} = 0.7\ \text{V}$
- $I_{CQ} = I_{EQ}$
- Bias-Stability Condition: $R_{TH} = 0.1(1+\beta)R_E$

Step 1: Choose R_E and Calculate I_{CQ}

A standard design rule for bias stability is to allocate approximately 10% of V_{CC} across R_E

$$V_{RE} \approx 0.1 * Vcc = 0.5\ V$$

Using Kirchhoff's Voltage Law (KVL) around the collector-emitter loop , Calculate I_{CQ}

$$V_{CC} = I_{CQ}R_C + V_{CEQ} + I_{EQ}R_E$$

I_{CQ} = 1.5 mA

Calculate R_E

$R_E = V_{RE} / I_{EQ}$

$= V_{RE} / I_{CQ}$

= 333.33 Ω

Step 2: Calculate I_{BQ}

$I_{BQ} = I_{CQ} / \beta$

$= 12.5\ \mu A$

Step 3: Calculate R_{TH}

For a bias-stable circuit, $R_{TH} = 0.1(1 + \beta)R_E$

Calculate R_{TH}

R_{TH} **= 4033.3 Ω**

Step 4: Calculate V_{TH}

Step 4: Calculate V_{TH}

Applying KVL at the base-emitter loop:

$$V_{TH} = I_{BQ}R_{TH} + V_{BE} + I_{EQ}R_E$$

$$V_{TH} = (0.0125\ \text{mA} \times 4.033\ \text{k}\Omega) + 0.7\ \text{V} + 0.5\ \text{V}$$

$$V_{TH} = 0.0504\ \text{V} + 0.7\ \text{V} + 0.5\ \text{V} = \mathbf{1.25\ V}$$

Step 5: Calculate R_1 and R_2

Step 5: Calculate R1 and R2

$$V_{TH} = Vcc \quad \frac{R_2}{R_1+R_2}$$

$$\frac{R_2}{R_1+R_2} = \quad ?$$

$$R_{TH} = \frac{R_1 R_2}{R_1+R_2}$$

$$= R_1 * \frac{R_2}{R_1+R_2}$$

R_1 = 16.13 kΩ $\qquad$ $R_2 = 5.38\ k\Omega$

Ex 5.18: Consider the circuit shown in Figure 5.62. Including bias resistors R_1 and R_2, even when bo
positive and negative bias sources are used, allows flexibility in the design. Let $\beta = 150$, $R_E = 0.2$ k
and $R_L = 1$ kΩ. Design a bias-stable circuit such that the quiescent output voltage is zero. (An
$R_1 = 16.7$ kΩ, $R_2 = 3.60$ kΩ)

